Clinical Features

Skin

Approximately one-half of patients with xeroderma pigmentosum (XP) have a history of acute sunburn reactions following minimal UV exposure, while the remaining patients tan normally without excessive burning. In all patients, numerous freckle-like hyperpigmented macules develop predominantly on sun-exposed skin (see Fig. 140-2). The median age of onset of cutaneous symptoms in XP is between 1 and 2 years (Fig. 140-3). These lesions typically spare sun-protected areas such as the buttocks, although severely sun-exposed individuals may exhibit pigmentary changes in the axillae.

With continued sun exposure, the skin becomes dry and parchment-like, with increased pigmentation—features that give rise to the name *xeroderma pigmentosum* ("dry pigmented skin"; see Fig. 140-2A). Premalignant actinic keratoses appear at an early age (see Fig. 140-2B). The appearance of sun-damaged skin in children with XP resembles that seen in adults such as farmers or sailors after decades of extreme sun exposure.

Cancer

Patients with XP under the age of 20 have a greater than 1000-fold increased risk of developing cutaneous basal cell carcinoma, squamous cell carcinoma, or melanoma.[11,12,15] Overall, there is an approximate 10- to 20-fold increase in internal neoplasms in XP patients.

The median age of onset of skin cancer in XP patients is dramatically reduced by approximately 50 years compared to the general U.S. population (Fig. 140-3).

Eyes

Ocular abnormalities are nearly as common as cutaneous findings and represent a key feature of XP (see Fig. 140-2C).[13,16] The retina is shielded from UV radiation by the anterior structures of the eye (eyelids, cornea, and conjunctiva), so clinical manifestations are largely confined to these UV-exposed anterior segments.

Photophobia is often present and may be associated with prominent conjunctival injection. Schirmer's testing frequently reveals reduced tear production, leading to dry eyes. Chronic UV exposure can result in severe keratitis, progressing to corneal opacification and neovascularization.

Eyelids show increased pigmentation, loss of eyelashes, and skin atrophy, which may lead to ectropion, entropion, or, in severe cases, complete eyelid loss. Benign conjunctival inflammatory masses or papillomas of the eyelids may occur. Malignancies such as epithelioma, squamous cell carcinoma, and melanoma commonly arise in UV-exposed ocular tissues. Ocular manifestations may be more severe in individuals with darker skin pigmentation.[19]

Neurologic System

Neurologic abnormalities occur in approximately 30% of XP patients.[12,15,18] Onset may occur in infancy or be delayed until the second decade of life. Neurologic involvement ranges from mild (e.g., isolated hyporeflexia) to severe, progressive conditions including mental retardation, sensorineural deafness (typically starting with high-frequency hearing loss), spasticity, and seizures.